



INTRO TO HIGH SECURITY HPC (RIO)

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High Performance Computing (HPC) at UVA

HPC computing systems tailored to data sensitivity

- Rivanna/Afton (Standard Security)
 - To process and store public data, internal use data, or sensitive data.
 - Highly Sensitive Data is not permissible
 - More information on the SSZ HPC can be found in our [Intro to HPC slides](#).
- Rio (High Security)
 - To process and store Highly Sensitive Data such as HIPAA, FERPA, Controlled Access Data (CAD), etc.
 - **Not** CUI certified yet, use Ivy VM

For more information, please see [RC Systems Data Sensitivity](#)

Highly Sensitive Data

Data that require restrictions on access under the law or that may be protected from release in accordance with applicable law or regulation.

Examples:

- Personally identifiable information (PII) is any information that can be used to identify a person. Examples include social security number, passport number, driver's license number, military identification number, or biometric records
- Health information that reveals an individual's health condition and/or medical history, including information defined by the Health Insurance Portability and Accountability Act (HIPAA)

Ivy and Rio (Overview)

Ivy

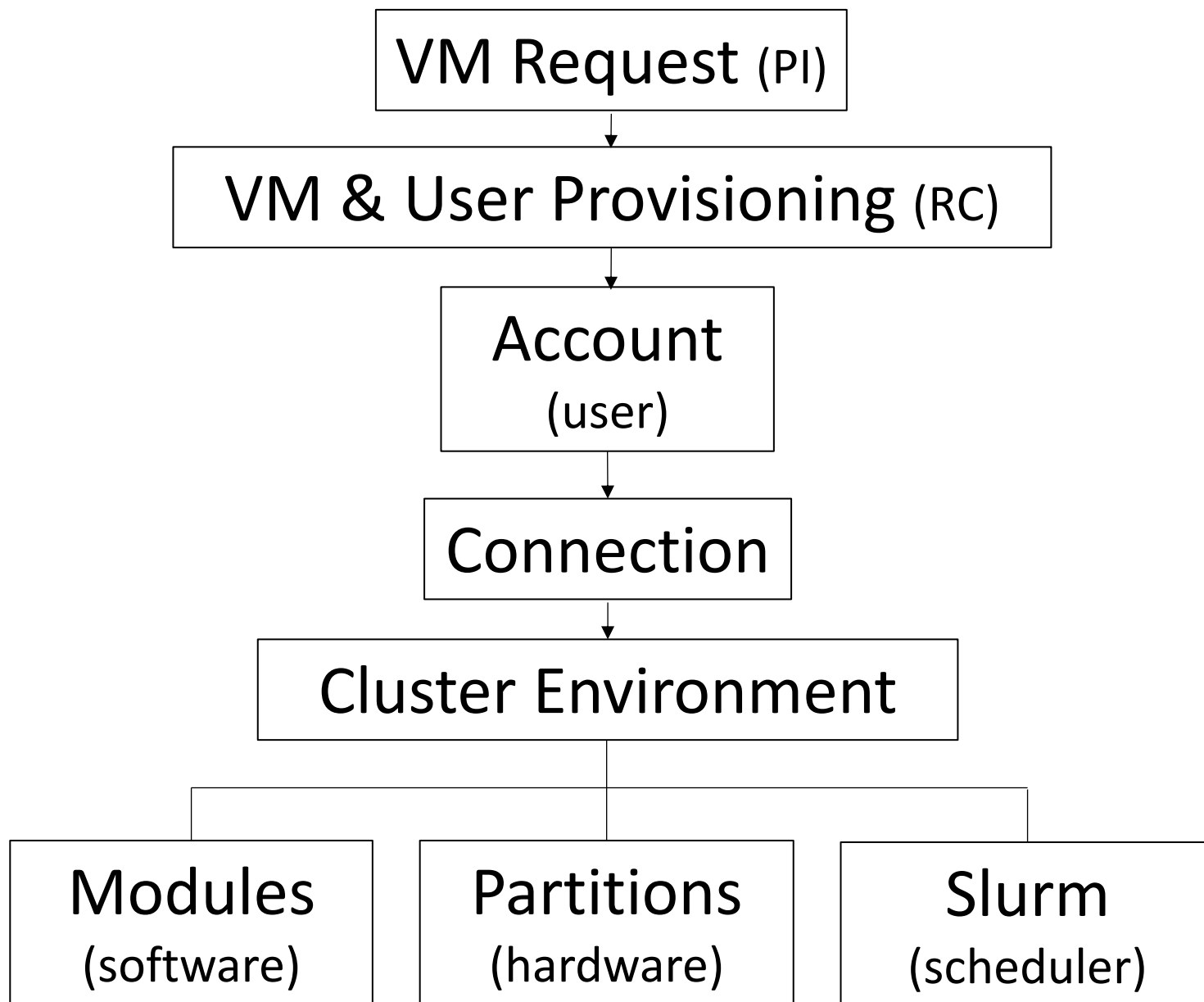
- Secure computing environment, offering Virtual Machines (VM) with both Linux and Windows operating systems.
- Ivy VM is designed for interactive and small-scale analysis of highly sensitive data, including HIPAA, FERPA, and CUI.
- Access to the Ivy platform is project based. Each project gets a VM.
- Project-specific storage volumes are mounted to the VM and available on the HPC system, enabling a seamless transition between tasks on VM and HPC environment.
- Ivy Linux VMs can serve as a frontend for accessing the Rio HPC system (available upon request).

Rio

- For large scale analysis of HIPAA, FERPA, and CAD

Terminology

- Node
 - Nodes are a type of computer called a server
 - They generally have more power than a typical computer.
 - They may have specialty hardware like graphical processing units
 - Two types of Nodes:
 - **Login node** - a server used for logging in and submitting jobs.
 - **Compute node** - a server that carries out the computational work.
- Core
 - An individual processor on a computer
- Memory
 - The random-access memory on a node
- Storage
 - Disk storage visible from a node



Requesting Access to HSZ HPC (Rio)

- Principal Investigators (PIs) can request access by submitting an Ivy Linux VM request through the [services app](#)
 - Research staff may apply for an exception.
 - Students must be sponsored by a faculty member or approved research staff.
 - In addition to being a login node to Rio, a Linux VM can be used for preliminary computations as well
 - VMs come in different sizes. Request a VM sized appropriately for your workflow. For larger groups or projects with computationally intensive tasks, choose a larger VM. We recommend Small and above.
 - VMs exist in a private, secure network and cannot reach outside resources on the Internet. Inbound and outbound data transfer is managed through the Data Transfer Node (DTN).
 - More information on requesting a VM can be found [here](#) on our website.

VM Provisioning

After access is requested, the Research Computing (RC) team will handle the provisioning of the Virtual Machine (VM).

The following steps are carried out by the RC team

- RC will create a Grouper group for Rio access
- RC will request access to the High Security VPN (HSVPN) on your group's behalf and provision 1 TB of High Security Research Standard storage for your project. This storage volume will also be available when using Rio HPC for your project.
- The PI and researchers will be notified upon the completion of the VM provisioning, and the associated IP address and group name will be provided to them.

User Provisioning

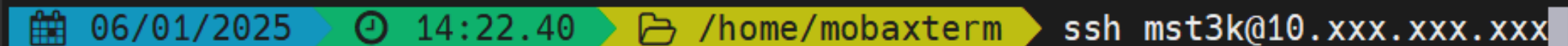
- Any users of the VM must be added to the project in the [services app](#) by the PI
- Once a project is approved, a PI and her/his researchers must sign a RUDA (one for every researcher on each project).
 - Each researcher must also complete the [High Security Awareness Training \(HSAT\)](#)
 - This must be completed to permit each researcher access to the HSVPN filter

Connecting to the System

- Before connecting to your VM, you must install the following on your workstation.
 - Cisco AnyConnect Secure Mobility Client.
 - Opswat.
 - Duo MFA on personal smartphone.
 - Details and links to each are on our [website](#).
- There are two ways to connect to your Linux VM:
 - Accessing the VM from a browser window
 - Using SSH (Secure Shell)
 - Use a terminal window with Mac/Linux
 - Windows users will need to install an SSH client application. We recommend [MobaXTerm](#).
 - In either case, you need the IP address of your VM along with your Netbadge credentials to connect to the VM.

Accessing from SSH

- Start the HSVPN
- Open your SSH client and type: `ssh mst3k@10.xxx.xxx.xxx`, where mst3k is replaced with your user ID and the x's are replaced with your VM's IP address (given in the Services app).

A screenshot of a terminal window with a dark background. The top bar shows a calendar icon, the date '06/01/2025', a clock icon, the time '14:22.40', a folder icon, and the path '/home/mobaxterm'. The main area shows the command 'ssh mst3k@10.xxx.xxx.xxx' with a cursor at the end.

```
06/01/2025 14:22.40 /home/mobaxterm ssh mst3k@10.xxx.xxx.xxx
```

- When prompted for a password, use your EServices password.

This login method provides only command-line access to the VM.
No graphical user interface (GUI) is usable.

SSH Clients

- SSH clients mostly use instructions typed at the command prompt.
- You will need to become familiar with Linux/Unix commands.
 - See our [Command Line Tutorial](#) to learn more
- Some useful Unix/Linux commands:

```
$ ls  
$ pwd  
$ cd folder_name  
$ cp file_1 file_2  
$ rm file_1  
$ cd ..
```

Accessing from a Browser

- Start the HSVPN
- Open a web browser and enter the IP address for your VM (e.g., <https://10.xxx.xxx.xxx>)
 - If you get a warning message, you may need to click on Advanced Settings and/or a Connect Anyway option, depending on your web browser.
- Logging in will take you to a web application called Open OnDemand (OOD).

VM Storage

Storage is split across two sets of directories:

1. Personal directory for individual data storage
 - Mounted under /home/\$USER
 - Little space is allotted for /home directories per VM
 - Individual home folders are personal, and not shareable
2. Shared storage space.
 - Mounted under */standard/ivy-hip-name* where “ivy-hip-name” is replaced by the name of your Ivy project’s Grouper group name
 - Default of 1TB of Research Standard Storage. PIs can get more when first requesting the Ivy project.
 - Storage can be resized using our [Storage Request Form](#).

Rio Caveat

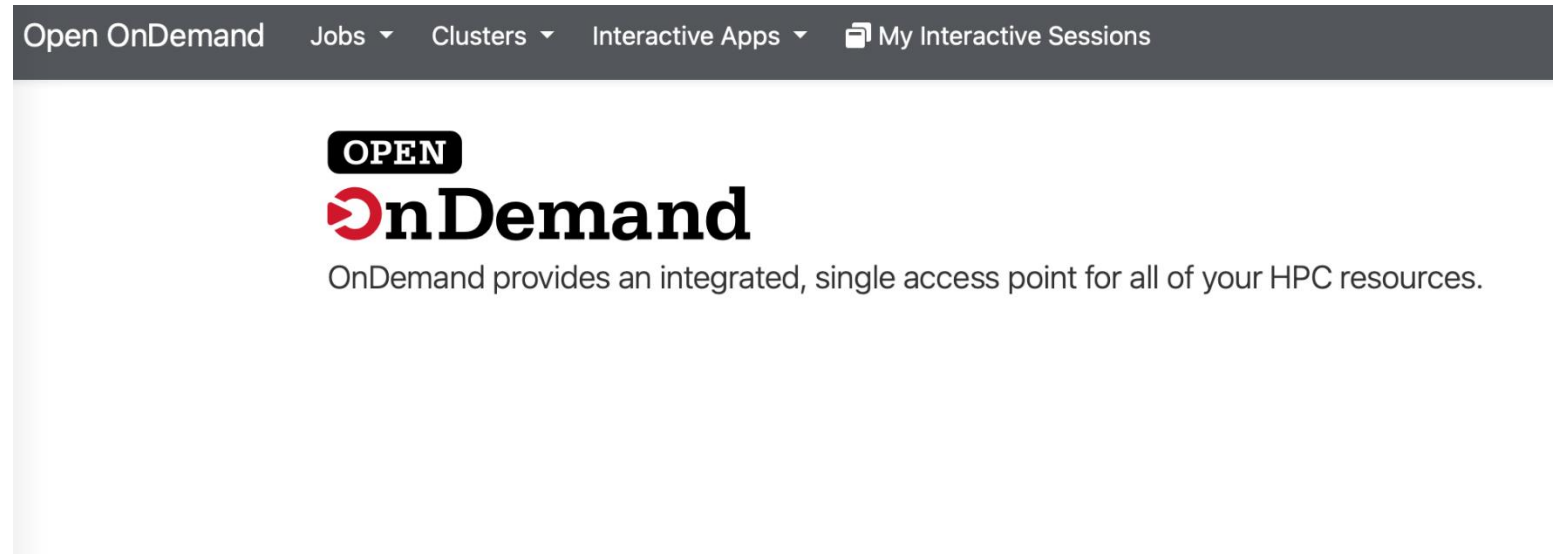
- One caveat to consider when working in the Rio environment is that /home is not mounted on Rio compute nodes.
 - This means that if your compute jobs references any files in your /home storage, they will not be found.
 - Ivy VMs not using Rio can ignore this condition.
- We recommend working exclusively out of the VM's Research Standard storage directory for Rio.
 - Individual user subdirectories are automatically created under /standard to organize the space.

Transferring Data

- Globus is the only permitted data-transfer protocol for highly sensitive data.
- ***UVA IVY-DTN*** is the official collection for moving files into High-Security Research Standard Storage.
 - To transfer data to High-Security Research Standard Storage, please see our [Globus instructions](#).
 - Ensure that you are **NOT** connected to the HSVPN. Data transfer will not work if you are connected to the HSVPN.
- Additional instructions and information on using Globus can be found on our [learning site](#).

Open On Demand

- Open OnDemand is a web application that allows for HPC access through a graphical browser window.
 - Features of the application are accessible using the top gray bar:
 - Jobs
 - Clusters
 - Interactive apps



Active Jobs

- Under the jobs dropdown is the active jobs tab
 - This shows all jobs currently running or queued on all partitions for all users.
 - You can filter the search for All Jobs or Your Jobs.
 - The Filter search bar allows you to narrow jobs by either user, queue, job name, job ID, or job status (running, queued, completed).
- More job details can be viewed using the dropdown to the left of a given job.
 - Completed jobs will be visible only for a short while as they are exiting.

Clusters

- The Clusters tab on the menu bar opens a new browser tab with a Linux command line interface for shell access.
 - This gives you terminal access to the frontend Ivy VM, similar to logging in via SSH.
 - Here, you can manipulate files, install software packages for Python & R, and submit command line jobs. Running processes here does not utilize Rio compute node resources.

Interactive Apps

- Open OnDemand offers GUI applications to interactively run jobs on Rio compute nodes.
- Job submissions are managed by a resource manager called Slurm.
- The following applications are available:
 - JupyterLab
 - Rstudio Server
 - FastX Desktop

Requesting a Session

- To request a session, select one of the applications from the dropdown. This will prompt you to fill in the desired resources for your interactive job.
- The example to the right is for starting a JupyterLab session.

JupyterLab

This app will launch [JupyterLab](#) on the [Rivanna/Afton cluster](#).

Rio Partition

Standard

Number of hours

1

Number of cores

1

Memory Request in GB

6

Allocation

hpc_build

Show Additional Options

No

Launch

* The JupyterLab session data for this session can be accessed under the [data root directory](#).

Choosing Resource Requests

- Partition – Standard or GPU (only select GPU if you are using GPU-enabled code)
- Time – Time limit for your session. Once reached, your session will terminate along with any running code. Cannot be adjusted.
- Cores – Used in parallel processing. Your code must be modified to take advantage of using multiple cores.
- Memory – A good rule of thumb is to request 2 to 3 times the size of the data that you are reading in or generating.
- Optional: Slurm Option – Used to input one of the many other Slurm options.

Queueing the Session

- Once you've filled out the resource request form, click the Launch button at the bottom to queue your job.
 - While queued, the job will wait for the resources you've asked for to become available.
 - Requests with higher resource requests (more cores, more memory, more time) may wait longer.

JupyterLab (173500) **Queued**

Created at: 2025-08-20 11:00:02 EDT ✕ Delete

Time Requested: 1 hour

Session ID: [5dc83b02-8550-4822-8ddf-9cfdd522c72b](#)

Please be patient as your job currently sits in queue. The wait time depends on the number of cores as well as time requested.

Launching the Session

- When the job has started, the status will change from Queued to Running. A button will appear to open a browser, launching the interactive session.

JupyterLab (173500)

1 node | 1 core | Running

Host: >_cdc-v15-3c1-s

Delete

Created at: 2025-08-20 11:00:02 EDT

Time Remaining: 59 minutes

Session ID: 5dc83b02-8550-4822-8ddf-9cfdd522c72b

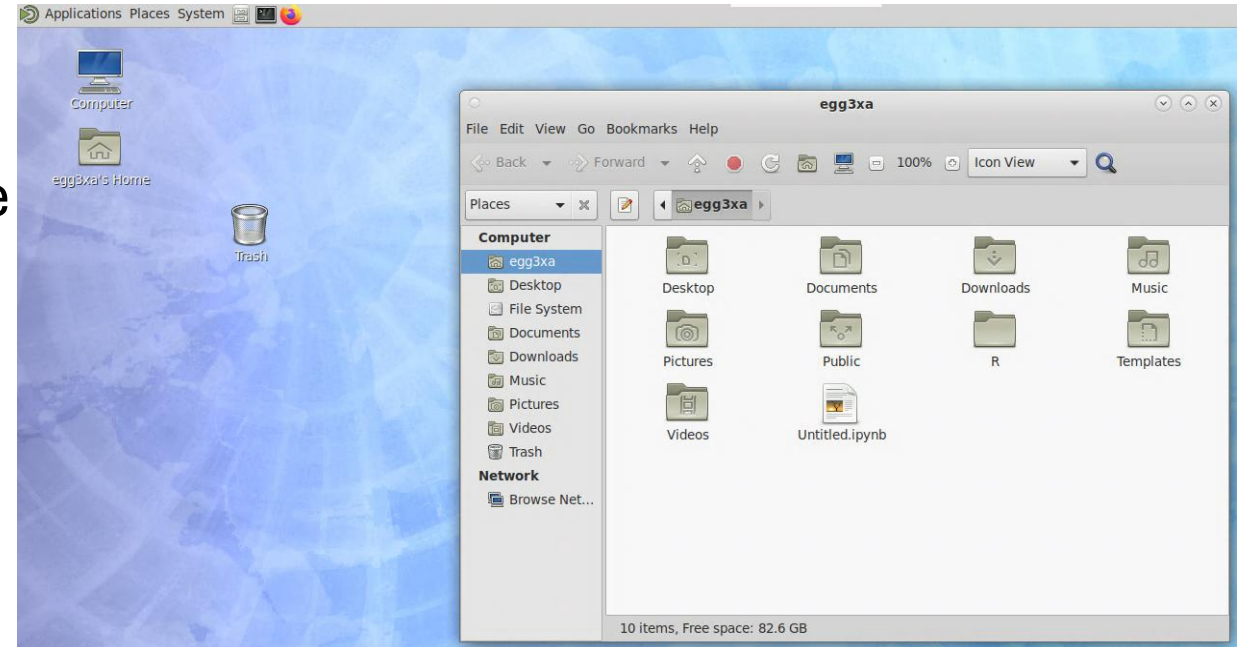
Connect to Jupyter

Interactive Desktop

- The Interactive Desktop application launches a virtual desktop session.
 - The Desktop app is a great tool for managing files and launching applications with a graphical user interface.
 - A Firefox browser is available and can be used to browse the web. The browser must be launched from a terminal with the network proxy set. Use the following commands in a terminal:
 - `export HTTPS_PROXY=http://figgis-s.hpc.virginia.edu:8080`
 - `export HTTP_PROXY=http://figgis-s.hpc.virginia.edu:8080`
 - `firefox`

Working with Files Using the Desktop

- The Desktop app has a graphical file manager that can be opened by clicking the folder icon on the desktop or the file cabinet icon on the applications bar.
- Using it should feel familiar. You can click and drag to move files/folders. Right-click to cut, copy, and paste.
- You can create and edit files by using the Pluma editor. Right-click in the empty space and select “Create Document” to create a text file. Right-click the new file and select “Open with Pluma” to edit it. You can edit existing files this way.



MATE Desktop with CAJA file browser open (white box on menu bar)

How to edit a file in Linux VM

Editors you can use:

- When in a Terminal Window, use vi/emacs/pluma.
 - vi/emacs uses all keyboard shortcuts and may take a while to learn.
 - pluma is a very basic editor. To launch, type: `pluma myfile.txt`
- In Desktop, you can use Pluma Text Editor.

Preinstalled Software

- Every VM comes with a [base installation](#) of software by default. New software can be installed upon request.
 - Software not already approved will need to undergo a security evaluation to determine if it is suitable for a High Security environment.
 - Python and R packages are available to users through the normal pip, conda, and CRAN library installation methods.
- Access to installed software is done using the **lmod** Module System.

Modules

- Any application software that you want to use will need to be loaded with the **module load** command.
- For example:
 - `module load apptainer/1.3.1`
- Modules are loaded only in your current shell. For repeated use, you must:
 - Load software every time you log out and back in
 - Load software every time you run a batch job on a compute node

Modules Basics

Some basic commands to use lmod:

- `module avail` – lists every module available to load
- `module key <keyword>` - searches for modules in a category specified
 - Example: `module key bio`
- `module spider <module-name>` - prints information about the software including different offered versions
- `module load <module-name>` - load the desired software
 - Use `module load <module-name/ver-number>` to load a specific version
- `module unload <module-name>` – unloads the desired software
- `module list` – prints all currently loaded modules
- `module purge` – unloads all modules

Python & R packages

- Since /home is not mounted on compute nodes, locally installed Python and R packages must be staged on /standard instead.
 - Both languages, by default, install to a folder in /home. Configuration changes are necessary to use Python and R in your compute jobs on Rio.
- Instructions for each are posted on our website:
 - [Conda environments](#)
 - [R libraries](#)
- When using interactive JupyterLab or RStudio, extra setup is not necessary.

Slurm

- Slurm manages the hardware resources on the cluster (e.g. compute nodes/cpu cores, compute memory, etc.).
- Slurm allows you to request resources within the cluster to run your code.
 - It is used for submitting jobs to compute nodes from an access point (Ivy VM).
 - Frontends are intended for editing, compiling, and submitting jobs
 - Production jobs go to the compute nodes through the resources manager.
- Slurm has [documentation](#) and we have [condensed notes](#) on our website:

Partitions (Queues)

- Rio has several partitions (or queues) for job submissions.
 - You will need to specify a partition when you submit a job.
 - To see the partitions that are available to you, type **qlist** at the command-line prompt.

Queue (partition)	Total Cores	Free Cores	Jobs Running	Jobs Pending	Time Limit
neo	960	836	0	0	7-00:00:00
neo-gpu	104	104	0	0	7-00:00:00
standard	1520	1200	0	0	7-00:00:00
gpu	96	96	0	0	3-00:00:00

- The neo partitions are exclusive to a specific group and are not for general users of Rio.

Slurm Script Example

- A Slurm script is a bash script with Slurm directives (#SBATCH) and command-line instructions for running your program.

```
#!/bin/bash
#SBATCH --nodes=1           #total number of nodes for the job
#SBATCH --cpus-per-task=1   #total number of cores for the job
#SBATCH --time=1-12:00:00   #amount of time for the whole job
#SBATCH --partition=standard #the queue/partition to run on
#SBATCH --account=ivy-hip-name #name of your Ivy group

module purge
module load goolf R          #load modules that my job needs
Rscript myProg.R             #command-line execution of my job
```

Submitting a Slurm Job

- To submit the Slurm command file to the queue, use the **sbatch** command at the command line prompt.
 - /home only exists on VM. Compute nodes do not have access to /home.
 - Job submission needs to be done from /standard/storage/
 - Recommended to include the complete file paths in the script
- For example, if the script on the previous slide is in a file named job_script.slurm, we can submit it as follows:

```
[jus2yw@ivy-tst-rc-1 ivy-tst-rc]$ sbatch job_script.slurm  
Submitted batch job 18316
```

Submitting an Interactive Slurm Job

- To submit an interactive SLURM job to the queue, use the **ijob** command at the command line prompt.
- For example, if you want to run an interactive application on a compute node in the standard queue using one cpu, we can submit it as follows:

```
-bash-4.1$ ijob -c 1 -p standard -A ivy-hip-name -t 06:00:00
salloc: Pending job allocation 21640112
salloc: job 21640112 queued and waiting for resources
salloc: job 21640112 has been allocated resources
salloc: Granted job allocation 21640112
srun: Step created for job 21640112
udc-aw34-21c0-teh1m$
```

Checking Queued Jobs

- To display the status of only your **active** jobs, type: **queue -u <your_user_id>**
- The queue command will show pending jobs and running jobs, but not failed, canceled or completed job.

```
-bash-4.2$ queue -u mst3k
```

JOBID	PARTITION	NAME	USER	ST	TIME	NODES	NODELIST (REASON)
18316	standard	job_sci	mst3k	R	1:45	1	udc-aw38-34-1

- A job's status is indicated by:
 - PD – pending, R – running, CG – exiting

Canceling a Job

- To delete a job from the queue, use the **scancel** command with the job ID number at the command line prompt:

```
-bash-4.2$ scancel 18316
```

- To cancel all your jobs, run this command:

```
-bash-4.2$ scancel -u $USER
```

Checking Job Status

- `scontrol` returns information for a running job
 - `scontrol show job <jobid>`
- `sacct` will return accounting information about completed and queued jobs. See [Slurm's documentation](#) for a full list of options.
 - Use the option `-o <options,list>` to customize the output
 - Use the option `-S <yyyy-mm-dd>` to inspect jobs finished after the given start date
 - Use the option `-j <jobid>` to inspect a particular job
- `seff` will return information about the utilization (called the “efficiency”) of core and memory.
 - `Seff <jobID>`

NEED HELP?

Research Computing Zoom Office Hours

Tuesdays: 3 pm – 5 pm
Thursdays: 10 am – noon

To connect to the Zoom sessions,
go to
<https://www.rc.virginia.edu/support/#office-hours> and click on the
“Join us via Zoom” button

Or, contact us through the forms at:

<https://www.rc.virginia.edu/support/>